

PSLE MATHEMATICS PRACTICE PAPER 1

Fractions, Decimals, Percentages & Ratios

Duration: 1 hour

Total Marks: 50

Name: _____

Date: _____

SECTION A: Short Answer Questions (20 marks, 15 minutes)

1. Convert 0.75 to a fraction in simplest form. (2 marks)

2. What is 25% of 80? (2 marks)

3. Express $\frac{3}{5}$ as a decimal. (2 marks)

4. Simplify the ratio 12:18 (2 marks)

5. If a book costs \$15 and is on sale for 20% off, what is the sale price? (3 marks)

6. Arrange these in ascending order: 0.6, $\frac{3}{5}$, 58% (2 marks)

7. John has 45 marbles. If 20% are red, how many red marbles does he have? (3 marks)

8. Express $\frac{7}{8}$ as a percentage. (2 marks)

SECTION B: Long Answer Questions (30 marks, 45 minutes)

9. Sarah spent $\frac{2}{5}$ of her pocket money on a book and $\frac{1}{4}$ on a pen. What fraction of her pocket money did she spend altogether? (4 marks)

10. A shop has 240 items. 15% are defective. How many items are not defective? (3 marks)

11. The ratio of boys to girls in a class is 3:4. If there are 12 boys, how many girls are there? (3 marks)

12. A car travels 150 km in 2.5 hours. Find its average speed in km/h. (3 marks)

13. Mei has \$200. She spends 35% on food and 20% on transport. How much money does she have left? (3 marks)

14. Three friends share \$84 in the ratio 2:3:4. How much does each friend get? (4 marks)

15. A dress costs \$120. During a sale, it is reduced by 30%. What is the new price? After another 10% discount on the new price, what is the final price? (5 marks)

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SOLUTIONS

SECTION A SOLUTIONS

Q1: Convert 0.75 to a fraction

$$0.75 = 75/100 = 3/4 \text{ (Divide both by 25)}$$

Q2: 25% of 80

$$25/100 \times 80 = 20$$

Q3: 3/5 as decimal

$$3 \div 5 = 0.6$$

Q4: Simplify 12:18

$$12:18 = 2:3 \text{ (Divide both by 6)}$$

Q5: Book discount

$$20\% \text{ of } \$15 = \$3, \text{ Sale price} = \$15 - \$3 = \$12$$

Q6: Arrange in order

$$0.6 = 60\%, 3/5 = 60\%, 58\% \rightarrow 58\%, 3/5, 0.6$$

Q7: Red marbles

$$20\% \text{ of } 45 = 20/100 \times 45 = 9 \text{ red marbles}$$

Q8: 7/8 as percentage

$$7/8 \times 100\% = 87.5\%$$

SECTION B SOLUTIONS

Q9: Fraction spent

$$2/5 + 1/4 = 8/20 + 5/20 = 13/20$$

Q10: Non-defective items

$$15\% \text{ of } 240 = 36 \text{ defective, } 240 - 36 = 204 \text{ not defective}$$

Q11: Boys to girls

$$\text{Boys:Girls} = 3:4, \text{ If 12 boys, then } 12 \div 3 = 4 \text{ per unit, Girls} = 4 \times 4 = 16$$

Q12: Average speed

Speed = $150 \text{ km} \div 2.5 \text{ hours} = 60 \text{ km/h}$

Q13: Money left

Food: 35% of \$200 = \$70, Transport: 20% of \$200 = \$40, Left: $\$200 - \$70 - \$40 = \90

Q14: Share \$84 in ratio 2:3:4

Total parts = $2+3+4 = 9$, Each part = $\$84 \div 9 = \$9.33\ldots$ Actually $\$84 \div 9 = \$28/3$. Friend 1: $\$56/3$, Friend 2: \$28, Friend 3: $\$112/9$. Let me recalculate: Total = 9 units, Friend 1: $2/9 \times \$84 = \18.67 , Friend 2: $3/9 \times \$84 = \28 , Friend 3: $4/9 \times \$84 = \37.33

Q15: Dress discount

Original: \$120, After 30%: $\$120 - 36 = \84 , After 10% on \$84: $\$84 - \$8.40 = \$75.60$

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